UNIT 9 · RAPID REVISION

Formula and Fact Sheet

Measurement systems and strain gauges

Item	Value
Measurement stages	Transducer → signal conditioning → output
Active vs passive transducer	Self-generating (thermocouple, piezo) vs needs excitation (strain gauge, LVDT, RTD)
Accuracy vs precision	Closeness to true value vs repeatability
Sensitivity	Output change ÷ input change
Hysteresis	Loading vs unloading difference
Systematic vs random error	Correct by calibration vs reduce by averaging
Gauge factor	GF = (ΔR/R)/ε — metal ≈ 2, semiconductor 20–40
Strain unit	Microstrain, με = 10^{-6}
Bridge balance	R1/R2 = R4/R3
Quarter-bridge output	V_out ≈ (V_ex/4)·GF·ε
Nyquist criterion	f_s > 2·f_max (else aliasing)
ADC resolution	2^n levels — 12-bit = 4096
Dynamic sensor damping	≈ 0.6–0.7 of critical
Long-distance analog transmission	4–20 mA current loop

Physical measurements

Item	Value
Thermocouple	Seebeck emf; type K most used (to ~1200 °C)
RTD	Platinum; R rises with T; Pt100 = 100 Ω at 0 °C
Thermistor	Semiconductor; R falls steeply with T (NTC)
Pyrometer	Non-contact, radiation
Bourdon gauge	C-tube uncoils under pressure
Brake power	BP = 2πNT/60,000
SPL	20 log(p/p_0); p_0 = 20 μPa
Bomb vs Junkers calorimeter	Solid/liquid vs gaseous fuels
Gross vs net CV	Differ by the latent heat of the water formed
Flash point	Pensky–Martens / Abel
Viscosity	Redwood / Saybolt viscometer
Head-type flow meters	$Q = Cd \cdot A \cdot \sqrt{(2\Delta p/\rho)}$ — $Q \propto \sqrt{\Delta p}$
Venturi vs orifice	Low loss, costly vs cheap, high loss
Mass-flow meter	Coriolis

Measurement systems, strain gauges and DAQ (Topics 1–4)

- 3 stages: transducer → signal conditioning → output. Sensor detects; transducer converts to electrical. Active (thermocouple/piezo, self-generating) vs passive (strain gauge/LVDT/RTD, need excitation).
- Static: accuracy (true value) vs precision (repeatability), sensitivity, resolution, hysteresis, linearity. Dynamic: time constant, natural frequency, damping. Systematic (bias, correctable) vs random (averaging).
- Strain gauge: GF = (ΔR/R)/ε; metal ≈ 2, semiconductor 20–40; strain in microstrain (με); Wheatstone bridge quarter/half/full; balance R1/R2 = R4/R3; dummy gauge = temperature compensation.
- Signal conditioning = amplify + filter + bridge + linearize; transmit via 4–20 mA loop or telemetry/slip rings. DAQ: ADC, resolution 2^n levels.
- NYQUIST: $f_s > 2 \cdot f_{max}$ to avoid ALIASING (anti-alias filter). Dynamic sensing needs natural frequency ≫ signal frequency, damping ~0.6–0.7.

Physical measurements, fuel properties and flow (Topics 5–7)

- Temperature: thermocouple = Seebeck emf (active, wide range, K most common); RTD = platinum, R ↑ with T (accurate, linear); thermistor = semiconductor, R ↓ with T (sensitive, non-linear); pyrometer = non-contact radiation (high T).
- Pressure: manometer (low p), Bourdon C-tube (mechanical gauge), diaphragm/bellows, piezo (dynamic). Force = load cell/proving ring; torque = strain-gauge shaft; power = dynamometer, BP = 2πNT/60,000.

- Sound: sound level meter → dB(A), SPL = 20 log(p/p_0), p_0 = 20 μPa. Vibration: piezoelectric accelerometer + FFT; displacement/velocity/acceleration.
- Fuel: bomb calorimeter (solid/liquid CV), Junkers (gas); GCV vs NCV; cetane/octane by CFR engine; flash point Pensky–Martens; viscosity Redwood/Saybolt.
- Flow: head meters $Q = Cd \cdot A \cdot \sqrt{(2\Delta p/\rho)}$ → $Q \propto \sqrt{\Delta p}$ (orifice cheap/lossy, venturi low-loss/costly, Pitot = velocity); rotameter (variable area, direct); PD meters totalise; Coriolis = mass flow.

Further reading

- Doebelin — Measurement Systems: Application and Design.
- Beckwith, Marangoni & Lienhard — Mechanical Measurements.
- Sawhney — Electrical and Electronic Measurements and Instrumentation.
- ASTM/IS fuel-testing methods (calorific value, flash point, viscosity, distillation).
- ISO 2631 / ISO 5349 (vibration); strain-gauge technical notes from gauge manufacturers.
